

ROB 545: Kinematics, Dynamics, & Control

Velocity kinematics

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Velocity kinematics

Using a minimal set of coordinates $x \in \mathbb{R}^m$ to represent the end-effector configuration, we can write the forward kinematics as a function of the set of joint variables $\theta \in \mathbb{R}^n$

$$x(t) = f(\theta(t))$$

By the chain rule, the time derivative at time t is

$$\begin{aligned}\dot{x}(t) &= \frac{\partial f(\theta)}{\partial \theta} \frac{d\theta(t)}{dt} = \frac{\partial f(\theta)}{\partial \theta} \dot{\theta} \\ &= J(\theta) \dot{\theta}\end{aligned}$$

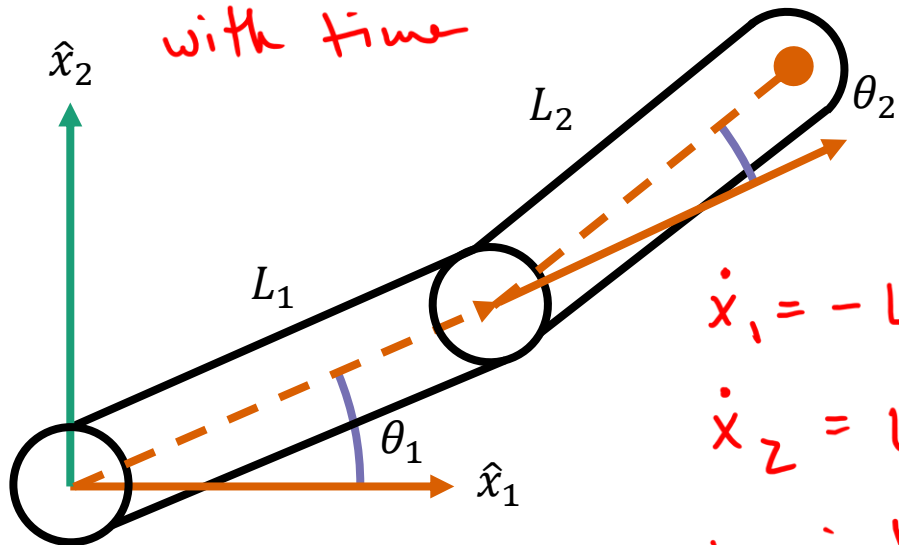
The Jacobian matrix: Mapping velocity

$J(\theta) \in \mathbb{R}^{m \times n}$ is the **Jacobian** matrix; it represents the linear sensitivity of the end-effector velocity $\dot{x} \in \mathbb{R}^m$ to joint velocity $\dot{\theta} \in \mathbb{R}^n$.

The Jacobian is a function of the joint variables θ ; that is, the resulting end-effector velocity depends on the current configuration of the robot.

Consider the planar 2R arm

* $x_1, x_2, \theta_1, \theta_2$ are variables that change with time (x_1, x_2)



F. K. of a 2R arm:

$$x_1 = L_1 \cos \theta_1 + L_2 \cos (\theta_1 + \theta_2)$$

$$x_2 = L_1 \sin \theta_1 + L_2 \sin (\theta_1 + \theta_2)$$

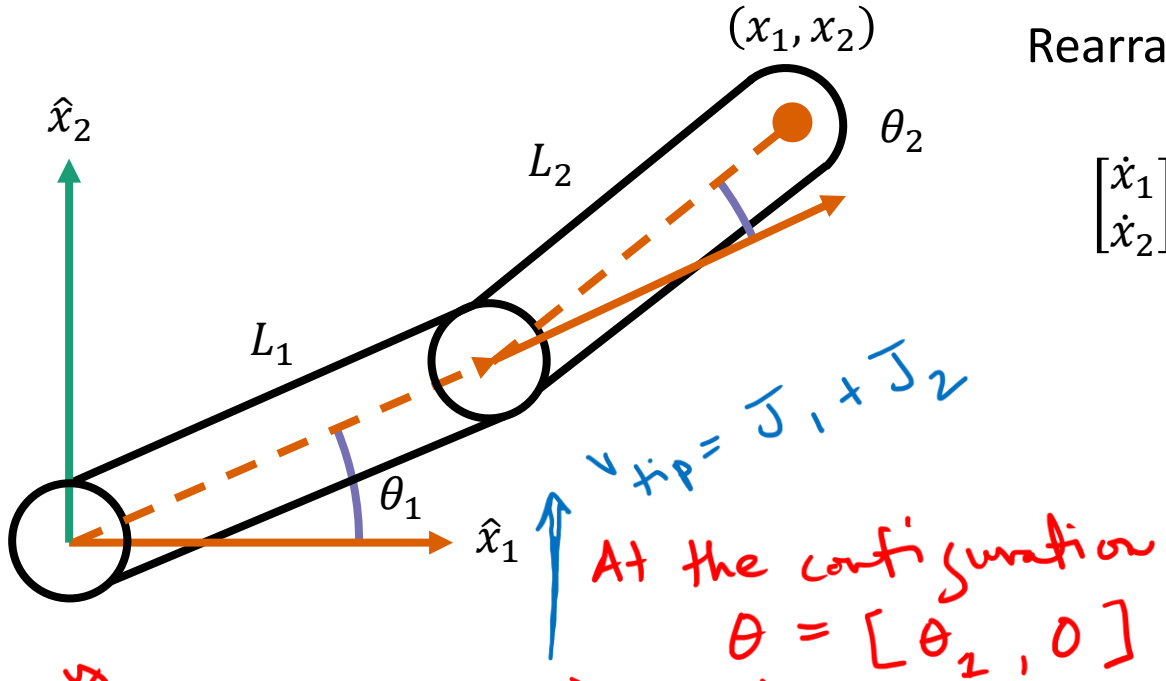
$$\dot{x}_1 = -L_1 \dot{\theta}_1 \sin \theta_1 - L_2 (\dot{\theta}_1 + \dot{\theta}_2) \sin (\theta_1 + \theta_2)$$

$$\dot{x}_2 = L_1 \dot{\theta}_1 \cos \theta_1 + L_2 (\dot{\theta}_1 + \dot{\theta}_2) \cos (\theta_1 + \theta_2)$$

Rearranging into the form $\dot{x} = J(\theta) \dot{\theta}$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -L_1 \sin \theta_1 - L_2 \sin (\theta_1 + \theta_2) & -L_2 \sin (\theta_1 + \theta_2) \\ L_1 \cos \theta_1 + L_2 \cos (\theta_1 + \theta_2) & L_2 \cos (\theta_1 + \theta_2) \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$\dot{x}$
 $J_1(\theta)$
 $J_2(\theta)$
 $\dot{\theta}$



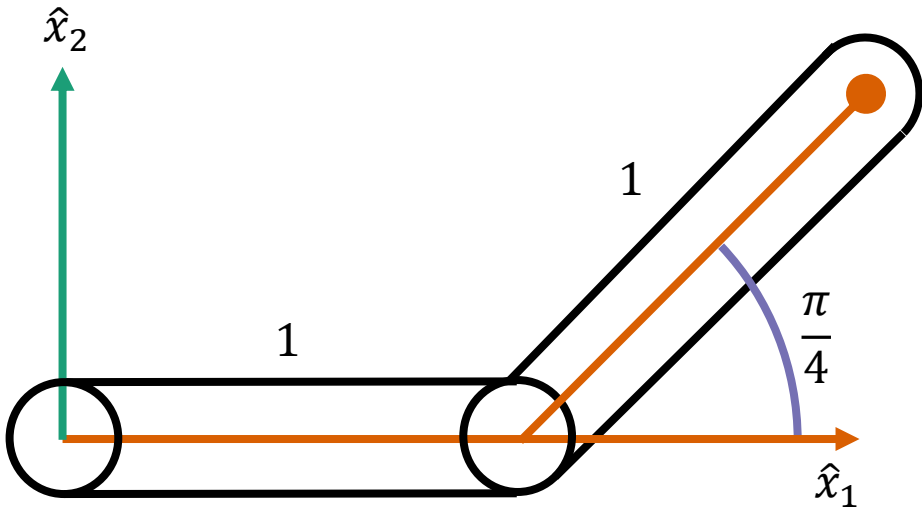
Rearranging into an equation of the form $\dot{x}(t) = J(\theta)\dot{\theta}$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2) \\ L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) \end{bmatrix}}_{J_1(\theta)} \underbrace{\begin{bmatrix} -L_2 \sin(\theta_1 + \theta_2) \\ +L_2 \cos(\theta_1 + \theta_2) \end{bmatrix}}_{J_2(\theta)} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$$\dot{x}(t) = v_{tip} = J_1(\theta)\dot{\theta}_1 + J_2(\theta)\dot{\theta}_2$$

We can choose $\dot{\theta}_1$ and $\dot{\theta}_2$ to generate v_{tip} in any arbitrary direction in the $x_1 - x_2$ plane.

This assumes that $J_1(\theta)$ and $J_2(\theta)$ are not collinear $\rightarrow$ Jacobian $J(\theta)$ is full rank, otherwise we have a **singularity**.

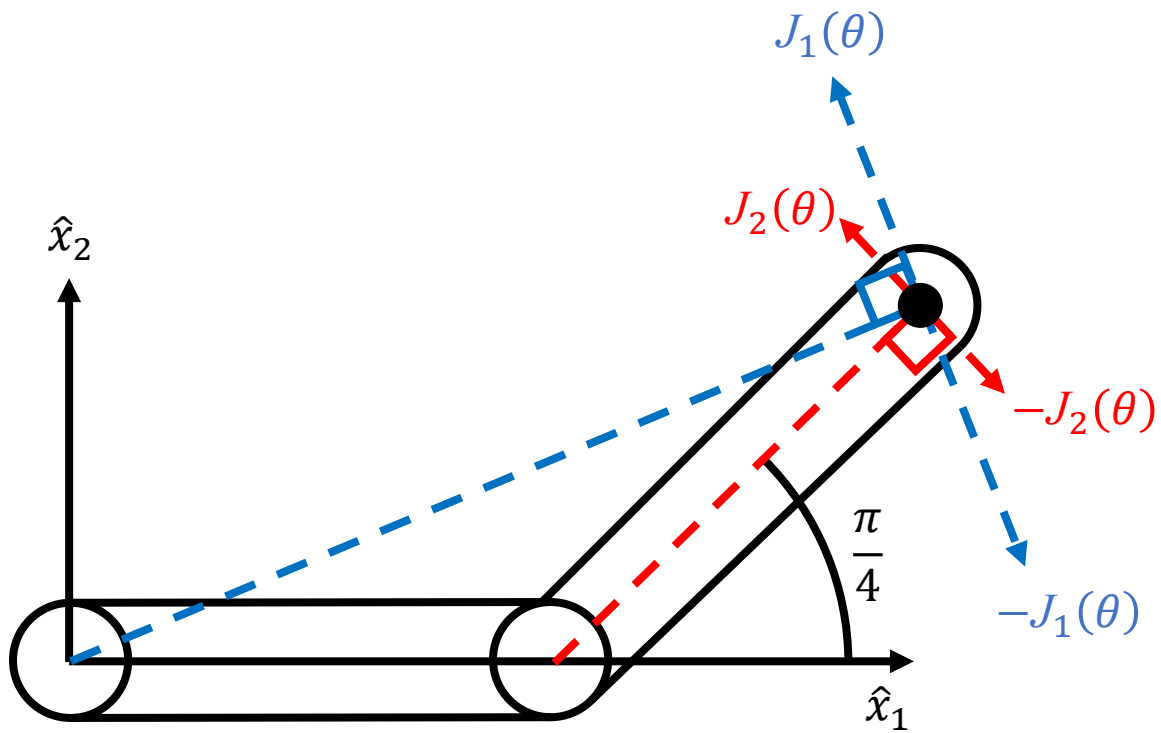


Let's say $L_1 = L_2 = 1$ and $\theta = (0, \pi/4)$

The Jacobian at this configuration is:

$$J\left(\begin{bmatrix} 0 \\ \pi/4 \end{bmatrix}\right) = \begin{bmatrix} -0.71 & -0.71 \\ 1.71 & 0.71 \end{bmatrix}$$

Each column of the Jacobian $J_i(\theta)$ corresponds to the tip velocity vector when $\dot{\theta}_i = 1$ and the other joint velocity is 0.



We can visualize the tip velocities associated with each column.

$$J\left(\begin{bmatrix} 0 \\ \pi/4 \end{bmatrix}\right) = \begin{bmatrix} -0.71 & -0.71 \\ 1.71 & 0.71 \end{bmatrix}$$

General form

For the general case, v_{tip} is a six-dimensional twist $\mathcal{V}$

- $J_i(\theta)$ is the twist $\mathcal{V}$ when $\dot{\theta}_i = 1$ and all other joint velocities $\dot{\theta}$ are zero
- The screw axes of the Jacobian depend on the current joint variables θ (for forward kinematics, the screw axes were always for the case $\theta = 0$)
- The space Jacobian $J_s(\theta)$ satisfies $\mathcal{V}_s = J_s(\theta)\dot{\theta}$ where each column $J_{si}(\theta)$ corresponds to a screw axis expressed in the fixed space frame $\{s\}$
- The body Jacobian $J_b(\theta)$ satisfies $\mathcal{V}_b = J_b(\theta)\dot{\theta}$ where each column $J_{bi}(\theta)$ corresponds to a screw axis expressed in the body frame $\{b\}$

Review of twists

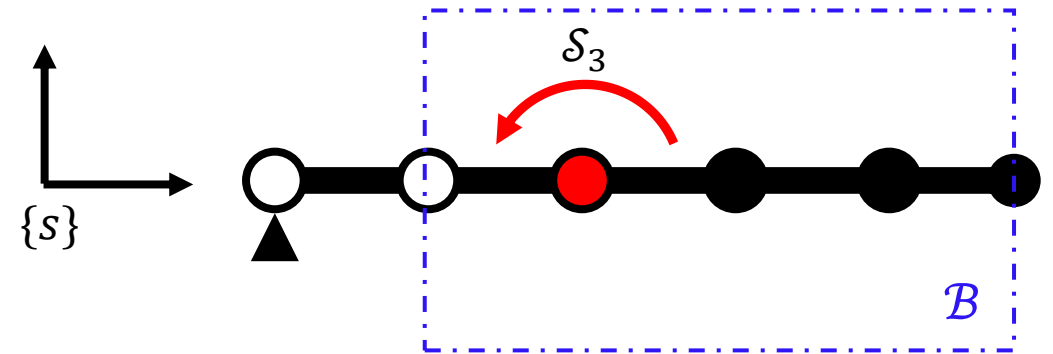
- $\mathcal{V}_s = [\omega_s, v_s]^T$ is the spatial twist, or the spatial velocity, in the space frame
 - ω_s is the angular velocity expressed in $\{s\}$
 - v_s is the linear velocity of a point at the origin of $\{s\}$ expressed in $\{s\}$
- $\mathcal{V}_b = [\omega_b, v_b]^T$ is the body twist, or the spatial velocity, in the body frame
 - ω_b is the angular velocity expressed in $\{b\}$
 - v_b is the linear velocity of a point at the origin of $\{b\}$ expressed in $\{b\}$

Visualizing the space Jacobian (1)

Consider the 5R planar arm to the right. Let $\mathcal{S}_3$ be the screw for joint 3 in the robot's zero configuration.

We know that $\theta_3, \theta_4, \theta_5$ won't affect the spatial twist from $\dot{\theta}_3$ since they won't displace axis 3 in $\{s\}$.

Fix those joint variables at zero, making the robot from joint 2 outward a rigid body $\mathcal{B}$.

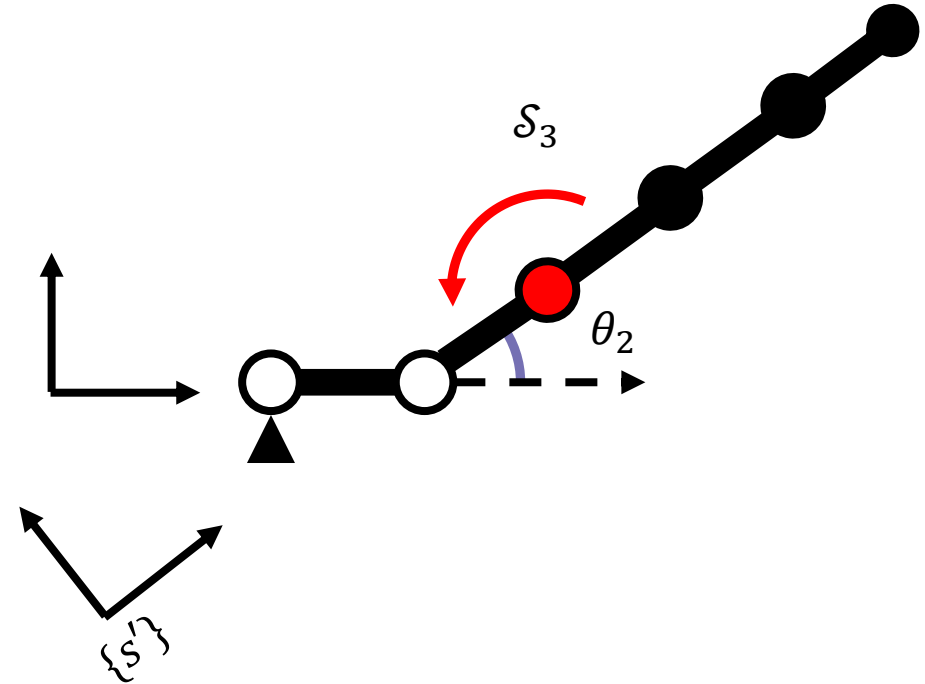


Visualizing the space Jacobian (2)

Now consider when $\theta_1 = 0$ and θ_2 is arbitrary.

Frame $\{s'\}$ is at the same position and orientation relative to $\mathcal{B}$ as frame $\{s\}$ when $\theta_1 = \theta_2 = 0$.

$$T_{ss'} = e^{[S_2]\theta_2}$$



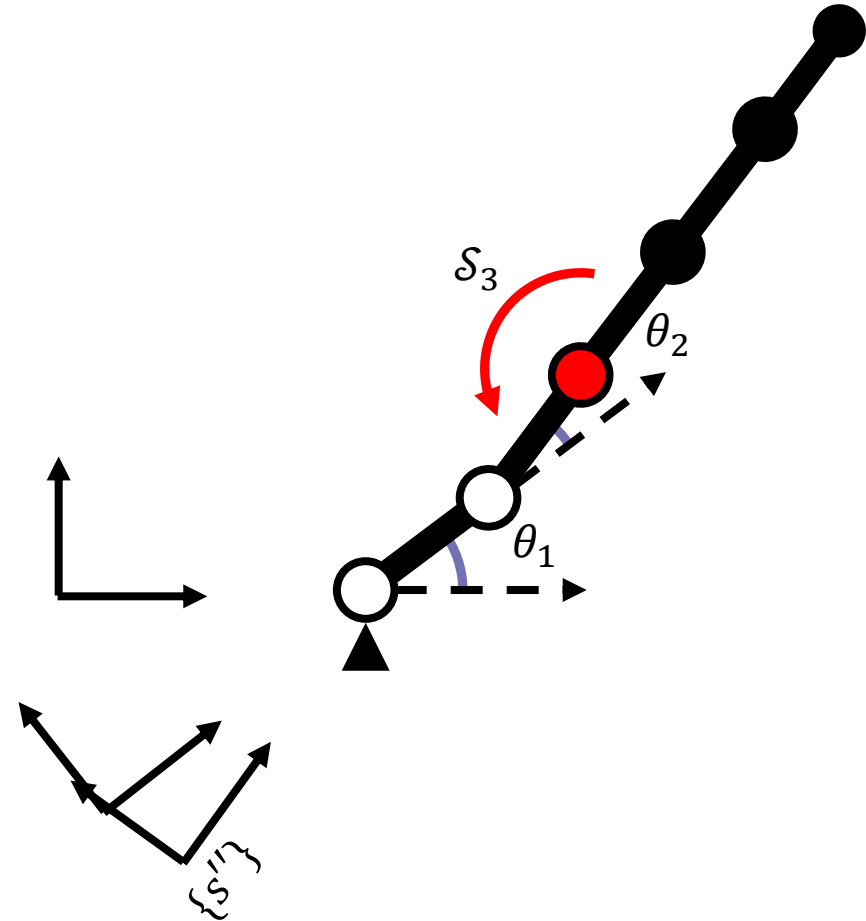
Visualizing the space Jacobian (3)

Now consider when both θ_1 and θ_2 are arbitrary.

Frame $\{s''\}$ is at the same position and orientation relative to $\mathcal{B}$ as frame $\{s\}$ when $\theta_1 = \theta_2 = 0$.

$$T_{ss''} = e^{[s_1]\theta_1} e^{[s_2]\theta_2}$$

So, $\mathcal{S}_3$ is the screw relative to $\{s''\}$ for arbitrary joint angles θ_1 and θ_2 .



Visualizing the space Jacobian (4)

If we look at the space Jacobian though, the column J_{S3} is the screw relative to $\{s\}$.
How can we relate $\mathcal{S}_3$ to J_{S3} ?

We can use adjoint mapping:

$$\begin{aligned} \left[\text{Ad}_{T_{ss''}} \right] &= \left[\text{Ad}_{e^{[S_1]\theta_1} e^{[S_2]\theta_2}} \right] \\ J_{S3} &= \left[\text{Ad}_{T_{ss''}} \right] \mathcal{S}_3 \end{aligned}$$

This way, we can change the representation of $\mathcal{S}_3$ from $\{s''\}$ to $\{s\}$.